

Literal Zero-Mode Disintegration: Constant-Mode Identities, Conditional-Expectation Normalization Barriers, and the Fixed-Shift Affine Correlation Gate

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Abstract

Let $h_0 \neq 0$ be fixed. In the matched prime-Möbius cutoff packet of TPC-32, the post-aggregation normalized-determinant coefficient at the physical content cutoff $C = \lfloor J \rfloor$ is

$$A_C(n), \quad Z_X = \sum_n A_C(n), \quad D_X = \sum_n |A_C(n)|^2.$$

The off-diagonal row mask gives $A_C(0) = 0$, but this coefficient identity does not imply that the finite Fourier value $\hat{A}_{C,q}(0) = Z_X$ vanishes. We give an exact disintegration of that distinguished Fourier mode and prove three normalization barriers.

First, under the strong physical no-wrap padding,

$$A_C = A_C^\circ + \frac{Z_X}{q} \mathbf{1}, \quad D_X = \|A_C^\circ\|_2^2 + \frac{|Z_X|^2}{q}.$$

Fourier inversion at the deleted determinant bin also gives

$$Z_X = - \sum_{r \neq 0} \hat{A}_{C,q}(r).$$

Thus canonical centering alone extracts the unknown constant mode rather than estimates it, while a coherent estimate for all nonzero modes is an exact alternative interface. Second, for a positive raw atomic measure whose determinant-fiber masses are M_n , conditional expectation onto the determinant sigma-algebra has squared norm

$$\sum_n \frac{|A_C(n)|^2}{M_n},$$

not D_X , and preserves the total integral Z_X . Unit mass per bin is therefore a change of normalization, not a consequence of conditional expectation. Third, exact common-divisor inversion does not center the packet: its $t = 1$ term is the unrestricted full matched shell and its $t > 1$ terms reassemble to the negative large-content shell.

We then trace the literal affine interface

$$\mu(d_0 + sr)\mu(u_0 + ar), \quad su_0 - ad_0 = h_0,$$

including growing slopes, short fibers, joint masks, and outer reassembly. This interface expands the raw hard coefficient; the calibrated drift remains a separate downstream ledger item. A quantifier audit shows why fixed-form logarithmic correlation theorems, shift averages, almost-all-scale results, and almost-all short intervals do not by themselves control this unweighted fixed- h_0 packet. An explicit dyadic sign sequence proves that logarithmic cancellation cannot be converted to ordinary cancellation by a structure-free Tauberian black box. The example is not multiplicative and asserts no obstruction to an argument using Möbius-specific structure.

All new finite identities and the explicit abstract countermodel are L0 results, and their placement inside the physical packet is an L1 transfer diagnosis. No new L2 affine Möbius estimate, parity-breaking theorem, prime-pair lower bound, or twin-prime conclusion is claimed.

Keywords: distinguished zero mode; determinant coefficient; conditional expectation; normalization barrier; fixed-shift affine correlation; logarithmic average; sieve parity.

Convention. Every physical statement retains the same fixed nonzero integer shift h_0 , the literal matched-shell coefficient, the actual determinant frequency $r = 0$, the content cutoff $C = \lfloor J \rfloor$, complete native keys, and one global post-aggregation normalization. The symbols A_C, Z_X, D_X refer throughout to the raw hard matched coefficient; calibrated drift legs are returned only in the downstream physical ledger. No averaged-shift, almost-all-scale, frozen-family, or bin-renormalized substitute is identified with the physical packet.

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1 Introduction

The determinant-frequency route of TPC-32 separates a literal signed arithmetic packet into coefficients $A_C(n)$ indexed by a normalized row determinant [9]. Parseval and the additive large sieve control the family of determinant frequencies in mean, but the physical reassembly selects one distinguished member:

$$\hat{A}_{C,q}(0) = \sum_n A_C(n).$$

An exceptional-frequency theorem is permitted to place this one frequency in its exceptional set. TPC-81 and TPC-82 therefore retain the bound for this zero mode as an independent arithmetic premise [11, 12].

This paper asks whether three natural reorganizations can reduce that premise: subtracting a mean, conditioning raw atoms on determinant fibers, or expanding the small-content indicator by Möbius inversion. Each operation is useful, but each has an exact limitation. Centering subtracts the unknown zero mode itself. Conditional expectation preserves the total integral and changes the post-bin energy by a fiber-mass factor. Content inversion returns the full shell in its principal term. These are normalization statements, not heuristic objections.

1.1 The literal object and the scope of the audit

Throughout, $h_0 \in \mathbb{Z} \setminus \{0\}$ is one fixed integer. It is not averaged, allowed to vary with X , or specialized to 2. The content cutoff is always

$$C = \lfloor J \rfloor. \tag{1}$$

The coefficient A_C is formed from the complete signed matched atoms before any norm or Fourier estimate is taken. In particular, the actual row-pair mask, fixed residue factors, smooth weights, both polarizations of the rank-two cutoff shell, and all native keys remain inside the coefficient. The energy

$$D_X = \sum_n |A_C(n)|^2$$

is the physical post-aggregation energy, not raw atom energy, an unsigned majorant, or a sum of pre-bin squares. Here “physical post-aggregation” refers to the hard raw matched coefficient of TPC-32. The calibrated packet contains two additional drift legs. They are restored only after the raw estimate and are charged in the downstream physical ledger; they are not silently inserted into A_C , Z_X , or D_X .

The audit has three levels.

Definition 1.1 (Proof levels). (i) L0 consists of finite identities in Fourier analysis, conditional expectation, divisor inversion, and affine parametrization, together with explicit structure-free countermodels that make a logical nonimplication exact.

(ii) L1 consists of a literal transfer or obstruction at the same fixed h_0 , with all physical keys and the global normalization retained.

(iii) L2 is a new asymptotic arithmetic estimate for the actual growing family, strong enough to save a fixed power after complete reassembly.

The theorems proved here are L0; their crosswalk to Z_X is L1. No L2 estimate is proved.

1.2 Main results

The first result is an orthogonal constant-mode identity. After embedding the determinant support injectively into a cyclic group of order q using the strong physical padding,

$$A_C^\circ = A_C - \frac{Z_X}{q} \mathbf{1} \quad (2)$$

satisfies

$$D_X = \|A_C^\circ\|_2^2 + \frac{|Z_X|^2}{q}. \quad (3)$$

At the deleted determinant bin, Fourier inversion gives the additional exact identity

$$Z_X = - \sum_{r \neq 0} \hat{A}_{C,q}(r). \quad (4)$$

Thus centering or mean-square information alone does not estimate the removed constant coefficient, but a coherent estimate for the full nonzero-frequency sum can do so. A generic-frequency theorem must return its exceptional nonzero modes and the coherent reassembly; it need not separately state a pointwise theorem at $r = 0$.

The second result starts with a general finite positive atomic measure ρ , a signed density f , and a bin map ν . If

$$A(n) = \int_{\nu=n} f d\rho, \quad M_n = \rho(\nu = n),$$

then

$$\|\mathbb{E}_\rho(f \mid \sigma(\nu))\|_{L^2(\rho)}^2 = \sum_{M_n > 0} \frac{|A(n)|^2}{M_n}. \quad (5)$$

Moreover,

$$\int \mathbb{E}_\rho(f \mid \sigma(\nu)) d\rho = \int f d\rho = \sum_n A(n).$$

Thus conditioning neither produces D_X nor cancels Z_X .

The third result applies the cumulative content projector

$$\eta_{\leq C}(t) = \sum_{\substack{c|t \\ c \leq C}} \mu(t/c).$$

For a target gcd G ,

$$\sum_{\substack{t|G \\ t > 1}} \eta_{\leq C}(t) = -\mathbf{1}_{G > C}.$$

Consequently the $t = 1$ term is exactly the unrestricted full shell, and the remaining terms are exactly minus the large-content shell. At (1), the known large-content estimate only gives the already recorded $X^{o(1)}Q^2$ splice error [8, 9].

Finally, we trace the raw hard physical column reduction to two affine Möbius signs and compare its quantifiers with the available correlation theorems. The fixed-determinant equation is

$$su_0 - ad_0 = h_0, \quad a = \ell j v,$$

but s and a grow with X , some fibers have bounded length, and the column must be returned through a joint mask and an outer sum. Theorems for fixed affine forms, averaged shifts, almost-all scales, or almost-all interval origins do not automatically supply that package. Content completion belongs to this raw reassembly, whereas the calibrated drift is returned only afterward.

1.3 A black-box Tauberian boundary

We also give a deterministic sequence $c_n \in \{-1, 1\}$ such that, for every $1 \leq \omega(x) \leq x$ with $\omega(x) \rightarrow \infty$,

$$\sum_{x/\omega(x) < n \leq x} \frac{c_n}{n} = o(\log \omega(x)),$$

while $\sum_{n \leq X} c_n$ has size comparable to X along an infinite dyadic subsequence. This proves that no structure-free Tauberian implication converts a Tao-shaped logarithmic estimate into ordinary cancellation. Because the constructed sequence is not multiplicative, this statement does not rule out a conversion based on additional Möbius structure.

1.4 Organization

[Section 2](#) fixes the literal coefficient and the coefficient-value/Fourier-frequency distinction. [Section 3](#) proves the no-wrap constant-mode decomposition. [Section 4](#) proves the conditional-expectation barrier, and [section 5](#) proves the content-projector reassembly. [Section 6](#) records the exact affine interface. [Section 7](#) audits the available theorems, while [section 8](#) gives the Tauberian counterexample. The endpoint ledger and claim boundary appear in [sections 9](#) and [10](#).

2 The literal matched coefficient and its distinguished zero mode

We restate the minimum part of the TPC-32 packet needed for the normalization analysis. This prevents an abstract coefficient sequence from being silently substituted for the physical object.

2.1 Rows, targets, and complete matched atoms

Let the physical rows be

$$\alpha = (\ell_\alpha, d_\alpha), \quad m_\alpha = \ell_\alpha d_\alpha \asymp Q,$$

and similarly for γ . For an allowed orbit index j , set

$$N_\alpha(j) = m_\alpha j + h_0, \quad N_\gamma(j) = m_\gamma j + h_0. \quad (6)$$

The primitive support conditions inherited from the packet ensure that every common target divisor can be cancelled against j , so that it divides $m_\alpha - m_\gamma$ [[8](#), [9](#)].

For a cutoff Y , write the cumulative divisor prefix as

$$A_{m,Y}(j) = \sum_{\substack{w \leq Y \\ w | m_j + h_0}} b_R(w), \quad (7)$$

where b_R is the divisor coefficient fixed by the physical construction. The matched cutoff kernel is

$$\mathcal{K}_{\alpha,\gamma}^{\text{sh}}(j) = A_{m_\alpha, U_0}(j) A_{m_\gamma, U_0}(j) - A_{m_\alpha, T}(j) A_{m_\gamma, T}(j). \quad (8)$$

Put

$$C_m = A_{m, U_0} - A_{m, T}, \quad H_m = \frac{A_{m, U_0} + A_{m, T}}{2} = A_{m, T} + \frac{1}{2} C_m.$$

The three physical channels and the exact two-leg polarization are

$$\mathcal{K}_{\alpha,\gamma}^{\text{sh}} = A_{m_\alpha, T} C_{m_\gamma} + C_{m_\alpha} A_{m_\gamma, T} + C_{m_\alpha} C_{m_\gamma} \quad (9)$$

and

$$\boxed{\mathcal{K}_{\alpha,\gamma}^{\text{sh}} = C_{m_\alpha} H_{m_\gamma} + H_{m_\alpha} C_{m_\gamma}.} \quad (10)$$

Both terms in (10) are retained; none of the arguments below deletes a polarization or separates the three channels by a triangle inequality.

Let $\mathcal{I}_X$ denote the complete finite set of allowed triples $\omega = (\alpha, \gamma, j)$. Define

$$w_X(\omega) = \gamma_\alpha^{(1)} \gamma_\gamma^{(2)} \mathfrak{A}_{\alpha, \gamma}(j) \mathcal{K}_{\alpha, \gamma}^{\text{sh}}(j). \quad (11)$$

Here the two γ 's are the literal opened-row coefficients and $\mathfrak{A}_{\alpha, \gamma}(j)$ is the actual joint multiplier. It contains the off-diagonal row mask, the fixed residue conditions, and the physical smooth weights. Thus w_X is a signed or complex amplitude, not its absolute majorant.

Remark 2.1 (Calibration convention). The calibrated prefix used in TPC-32 moves an explicit mean term into two drift legs. Those legs are part of the literal matched-shell ledger. The operation does not assert that the determinant coefficient introduced below has zero total sum.

2.2 Content and normalized determinant

For $\omega = (\alpha, \gamma, j)$, put

$$c(\omega) = \gcd(N_\alpha(j), N_\gamma(j)), \quad \Delta^\#(\omega) = \frac{m_\alpha - m_\gamma}{c(\omega)}. \quad (12)$$

The primitive support makes $\Delta^\#(\omega)$ an integer whenever the off-diagonal atom is active. At the physical cutoff

$$C = \lfloor J \rfloor, \quad (13)$$

define

$$A_C(n) = \sum_{\omega \in \mathcal{I}_X} w_X(\omega) \mathbf{1}_{c(\omega) \leq C} \mathbf{1}_{\Delta^\#(\omega) = n}, \quad n \in \mathbb{Z}. \quad (14)$$

This is aggregation before squaring. The two physical quantities of interest are

$$Z_X = \sum_{n \in \mathbb{Z}} A_C(n), \quad D_X = \sum_{n \in \mathbb{Z}} |A_C(n)|^2. \quad (15)$$

The first is the distinguished zero mode; the second is the literal post-bin energy.

Proposition 2.2 (Two meanings of zero). *The off-diagonal row mask implies*

$$A_C(0) = 0. \quad (16)$$

For every auxiliary modulus q , however,

$$\hat{A}_{C, q}(0) := \sum_{n \in \mathbb{Z}} A_C(n) e_q(0) = Z_X, \quad (17)$$

and (16) gives no cancellation in (17).

Proof. If $\Delta^\#(\omega) = 0$, then $m_\alpha = m_\gamma$. Such an atom is deleted by the actual off-diagonal mask, proving (16). At Fourier frequency $r = 0$, the character $e_q(-rn)$ equals one for every determinant bin n , which gives (17). Deleting the single bin $n = 0$ does not force the sum over the remaining bins to vanish. $\square$

Remark 2.3 (Frequency typing). The $r = 0$ in (17) is the finite Fourier frequency of the normalized determinant. It is not an orbit-variable Poisson zero mode and not a zero mode of a centered divisor kernel.

2.3 The full-shell splice

Let

$$\mathcal{S}_{\text{full}}^{\text{sh}} = \sum_{\omega \in \mathcal{I}_X} w_X(\omega), \quad (18)$$

$$\mathcal{S}_{>C}^{\text{sh}} = \sum_{\omega \in \mathcal{I}_X} w_X(\omega) \mathbf{1}_{c(\omega) > C}. \quad (19)$$

Proposition 2.4 (Exact zero-mode/full-shell identity). *At the literal cutoff (13),*

$$Z_X = \mathcal{S}_{\text{full}}^{\text{sh}} - \mathcal{S}_{>C}^{\text{sh}}. \quad (20)$$

In the inherited physical range,

$$|\mathcal{S}_{>C}^{\text{sh}}| \ll_{\varepsilon, h_0, \mathcal{D}} X^\varepsilon \left(Q^2 + \frac{XQ}{C} \right) = X^{o(1)} Q^2. \quad (21)$$

Consequently, up to an $X^{o(1)} Q^2$ error, estimating Z_X is the same problem as estimating the complete matched shell.

Proof. Exact contents partition $\mathcal{I}_X$, so summing (14) over n gives the small-content shell. Its complement is (19), proving (20). The first inequality in (21) is the inherited large-content estimate [8, 9]. The physical scale relation $JQ = X^{1+o(1)}$, together with $C = \lfloor J \rfloor$, gives $XQ/C = X^{o(1)} Q^2$. $\square$

Corollary 2.5 (The coherent completion required from generic frequencies). *Under the strong physical no-wrap padding (22), the distinguished mode satisfies*

$$Z_X = - \sum_{r \neq 0} \hat{A}_{C,q}(r).$$

Consequently, a generic-frequency or mean-square theorem controls Z_X only after it also controls the coherent sum of the retained nonzero modes and every exceptional nonzero mode. A direct estimate at $r = 0$ is sufficient, but it is not logically necessary.

Proof. This is (32). Bounds for most individual modes do not estimate the displayed sum when the uncontrolled nonzero modes or the phase-coherent reassembly can carry the full mass. Conversely, any estimate for the complete coherent sum is already an estimate for Z_X , regardless of whether a separate pointwise theorem was stated at $r = 0$. $\square$

3 No-wrap constant-mode decomposition

The determinant support of the physical coefficient lies in

$$[-B_{\mathcal{D}}Q, B_{\mathcal{D}}Q] \cap \mathbb{Z}$$

for a fixed $B_{\mathcal{D}}$ [9]. We retain the strong physical padding used there and choose an integer q so that

$$q > 2B_{\mathcal{D}}Q + 2, \quad q \asymp Q. \quad (22)$$

Thus reduction modulo q is injective on the support and no nonzero determinant bin is congruent to zero modulo q . Define the padded cyclic coefficient

$$\tilde{A}_C(a) = \sum_{\substack{n \in \mathbb{Z} \\ n \equiv a \pmod{q}}} A_C(n), \quad a \in \mathbb{Z}/q\mathbb{Z}. \quad (23)$$

The no-wrap condition gives

$$\sum_{a \pmod{q}} |\tilde{A}_C(a)|^2 = D_X, \quad \sum_{a \pmod{q}} \tilde{A}_C(a) = Z_X, \quad \tilde{A}_C(0) = A_C(0) = 0. \quad (24)$$

3.1 The orthogonal projection

Equip $\mathbb{C}^{\mathbb{Z}/q\mathbb{Z}}$ with the unnormalized inner product

$$\langle f, g \rangle_q = \sum_{a \pmod{q}} f(a) \overline{g(a)}.$$

Let $\mathbf{1}_q$ be the constant-one vector and define

$$P_0 f = \frac{\langle f, \mathbf{1}_q \rangle_q}{q} \mathbf{1}_q, \quad P_{\perp} = I - P_0. \quad (25)$$

Theorem 3.1 (Exact constant-mode disintegration and coherent reassembly). *Set*

$$A_C^\circ = P_\perp \tilde{A}_C = \tilde{A}_C - \frac{Z_X}{q} \mathbf{1}_q. \quad (26)$$

Then

$$\sum_{a \in (q)} A_C^\circ(a) = 0, \quad (27)$$

$$\tilde{A}_C = A_C^\circ + \frac{Z_X}{q} \mathbf{1}_q, \quad (28)$$

$$D_X = \|A_C^\circ\|_{\ell^2(\mathbb{Z}/q\mathbb{Z})}^2 + \frac{|Z_X|^2}{q}. \quad (29)$$

For the unnormalized finite Fourier transform

$$\hat{f}(r) = \sum_{a \in (q)} f(a) e_q(-ra),$$

one has

$$\widehat{A_C^\circ}(0) = 0, \quad \widehat{A_C^\circ}(r) = \widehat{A}_{C,q}(r) \quad (r \neq 0), \quad (30)$$

and hence

$$\sum_{\substack{r \in (q) \\ r \neq 0}} |\widehat{A}_{C,q}(r)|^2 = qD_X - |Z_X|^2 = q\|A_C^\circ\|_2^2. \quad (31)$$

The deleted determinant bin also gives the exact coherent identity

$$Z_X = - \sum_{\substack{r \in (q) \\ r \neq 0}} \widehat{A}_{C,q}(r). \quad (32)$$

Proof. By (24),

$$P_0 \tilde{A}_C = \frac{Z_X}{q} \mathbf{1}_q.$$

The ranges of P_0 and $P_\perp$ are orthogonal, while $\|\mathbf{1}_q\|_2^2 = q$. Pythagoras therefore gives (29). The Fourier transform of $(Z_X/q)\mathbf{1}_q$ equals Z_X at $r = 0$ and zero otherwise. This proves (30). Finite Parseval then gives (31). Finally, Fourier inversion at the cyclic coordinate $a = 0$, together with $\tilde{A}_C(0) = 0$, gives

$$0 = \frac{1}{q} \left(Z_X + \sum_{\substack{r \in (q) \\ r \neq 0}} \widehat{A}_{C,q}(r) \right),$$

which is (32). □

Corollary 3.2 (What centering alone does and does not prove). *Let*

$$S_X = \# \text{supp } A_C.$$

For the literal coefficient,

$$|Z_X|^2 \leq S_X D_X \leq (q-1) D_X \leq q D_X. \quad (33)$$

The final qD_X bound is the projection bound for an unrestricted cyclic vector, but it is not sharp for the physically padded coefficient because the zero coordinate is absent. The centering identity alone produces no fixed-power improvement over (33). On the other hand, (32) shows that a genuinely coherent estimate for all nonzero Fourier modes can estimate Z_X ; merely declaring the centered zero Fourier mode to vanish cannot.

Proof. Cauchy–Schwarz on the exact support gives

$$|Z_X|^2 = \left| \sum_{a \in \text{supp } \tilde{A}_C} \tilde{A}_C(a) \right|^2 \leq S_X D_X.$$

The strong padding and $\tilde{A}_C(0) = 0$ give $S_X \leq q - 1$. The last assertions follow from [equations \(29\) and \(32\)](#). $\square$

Remark 3.3 (Dependence on the ambient group). The decomposition is canonical after the physical no-wrap modulus q has been chosen. Centering only on the set $\text{supp } A_C$ is not an invariant replacement: exact cancellations can change that support, and adding zero bins changes its average. Different admissible no-wrap values of q give different constant vectors, although the physical sum Z_X itself is independent of q .

Remark 3.4 (Weaker no-wrap padding). If one assumes only $q > \text{diam}(\text{supp } A_C)$, reduction remains injective but a nonzero integer bin can lie in residue class zero. In that case Fourier inversion gives the more general identity

$$Z_X = q\tilde{A}_C(0) - \sum_{\substack{r \ (q) \\ r \neq 0}} \hat{A}_{C,q}(r).$$

The strong physical choice [\(22\)](#) is what identifies $\tilde{A}_C(0)$ with the deleted literal bin $A_C(0) = 0$.

Example 3.5 (Deleting the bin $n = 0$ is not centering). *Fix any $z \in \mathbb{C}$ and take an abstract coefficient supported at $n = 1$ with $A(1) = z$. Then $A(0) = 0$, but $\tilde{A}(0) = z$ and $D = |z|^2$. This finite example is not a model of the physical arithmetic; it isolates the logical nonimplication between the two zero statements in [proposition 2.2](#).*

4 Conditional expectation and the fiber-mass barrier

A second possible route is to regard determinant aggregation as a conditional expectation. This is exact only after the underlying atomic measure is specified. The resulting norm carries the masses of the determinant fibers.

4.1 A finite atomic theorem

Let Ω be a finite set, let ρ be a positive measure with $\rho(\{\omega\}) > 0$ for every $\omega \in \Omega$, and let

$$\nu : \Omega \longrightarrow \mathcal{B}$$

be a finite bin map. For $n \in \nu(\Omega)$, define

$$\Omega_n = \nu^{-1}(\{n\}), \quad M_n = \rho(\Omega_n), \quad A(n) = \int_{\Omega_n} f \, d\rho, \quad (34)$$

where $f : \Omega \rightarrow \mathbb{C}$. Put

$$Z = \sum_n A(n) = \int_{\Omega} f \, d\rho, \quad D_{\text{bin}} = \sum_n |A(n)|^2. \quad (35)$$

Theorem 4.1 (Conditional-expectation normalization identity). *Let $\mathcal{G} = \sigma(\nu)$. On Ω_n ,*

$$\mathbb{E}_{\rho}(f \mid \mathcal{G}) = \frac{A(n)}{M_n}. \quad (36)$$

Consequently,

$$\|\mathbb{E}_{\rho}(f \mid \mathcal{G})\|_{L^2(\rho)}^2 = \sum_n \frac{|A(n)|^2}{M_n}, \quad (37)$$

$$\int_{\Omega} \mathbb{E}_{\rho}(f \mid \mathcal{G}) \, d\rho = \int_{\Omega} f \, d\rho = Z, \quad (38)$$

$$\|f\|_{L^2(\rho)}^2 = \|\mathbb{E}_{\rho}(f \mid \mathcal{G})\|_{L^2(\rho)}^2 + \|f - \mathbb{E}_{\rho}(f \mid \mathcal{G})\|_{L^2(\rho)}^2. \quad (39)$$

In particular, conditional expectation naturally controls the fiber-normalized energy in (37), not D_{bin} .

Proof. A $\mathcal{G}$ -measurable function is constant on each nonempty fiber. The defining integral identity for conditional expectation forces that constant on Ω_n to be

$$\frac{1}{\rho(\Omega_n)} \int_{\Omega_n} f d\rho = \frac{A(n)}{M_n}.$$

Integrating its squared modulus gives

$$\sum_n M_n \left| \frac{A(n)}{M_n} \right|^2 = \sum_n \frac{|A(n)|^2}{M_n},$$

which proves (37). Integrating without the square proves (38). Finally, conditional expectation is the orthogonal projection from $L^2(\rho)$ onto $L^2(\mathcal{G}, \rho)$, giving (39). $\square$

Corollary 4.2 (Comparison requires fiber-mass bounds). *If*

$$0 < M_{\min} \leq M_n \leq M_{\max} < \infty$$

on the active bins, then

$$M_{\min} \|\mathbb{E}_\rho(f \mid \mathcal{G})\|_2^2 \leq D_{\text{bin}} \leq M_{\max} \|\mathbb{E}_\rho(f \mid \mathcal{G})\|_2^2. \quad (40)$$

Without quantitative control of $M_{\min}$ and $M_{\max}$, the two energies are not uniformly equivalent.

Proof. Write

$$D_{\text{bin}} = \sum_n M_n \frac{|A(n)|^2}{M_n}$$

and bound M_n above and below. $\square$

4.2 Application to the determinant map

Take

$$\Omega = \{\omega \in \mathcal{I}_X : c(\omega) \leq C\}, \quad \nu(\omega) = \Delta^\#(\omega).$$

With counting measure and $f(\omega) = w_X(\omega)$, (14) becomes

$$A(n) = A_C(n).$$

More generally, a preassigned positive native atom measure can be used by absorbing its density into f . Once that measure is chosen, it must remain the same through all bins if one global raw normalizer is to be retained.

Proposition 4.3 (Literal determinant conditioning). *For the literal determinant map,*

$$\|\mathbb{E}(f \mid \sigma(\Delta^\#))\|_2^2 = \sum_n \frac{|A_C(n)|^2}{M_n}, \quad \int \mathbb{E}(f \mid \sigma(\Delta^\#)) = Z_X. \quad (41)$$

Therefore:

- (a) *the conditional-expectation contraction from raw atomic energy does not by itself prove a lower bound for D_X ;*
- (b) *conditional expectation cannot create cancellation of Z_X ;*
- (c) *replacing the pushforward mass M_n by unit mass in every bin is a new normalization, not an identity.*

Proof. The two identities are [equations \(37\)](#) and [\(38\)](#). The contraction controls the left side of the first identity, while the physical energy is

$$D_X = \sum_n |A_C(n)|^2.$$

These differ by the factors M_n . The second identity proves part (b). Finally, the pushforward measure $\nu_*\rho$ assigns mass M_n to bin n . Replacing it by counting measure assigns mass one and therefore reweights bin n by M_n^{-1} , proving part (c). $\square$

Remark 4.4 (Centering after conditioning). Subtracting the global mean

$$f^\circ = f - \frac{Z_X}{\rho(\Omega)}$$

does give $\int f^\circ d\rho = 0$, but it uses the unknown Z_X . Its aggregate in bin n is

$$A_C(n) - \frac{M_n}{\rho(\Omega)} Z_X,$$

which is not the cyclic centering $A_C(n) - Z_X/q$ unless an additional equal-fiber-mass identity is proved. The two centering procedures must not be conflated.

4.3 Sharp finite examples

Example 4.5 (A single large fiber). *Let $\Omega = \{1, \dots, M\}$, give each atom unit mass, put all atoms in one bin, and take $f \equiv 1$. Then*

$$Z = M, \quad D_{\text{bin}} = M^2, \quad \|\mathbb{E}(f \mid \mathcal{G})\|_2^2 = M.$$

Thus the conditional-expectation energy and post-bin energy differ by the full fiber mass.

Example 4.6 (Raw energy can die inside a bin). *Place two unit-mass atoms in one bin and give them values 1 and -1 . Their raw L^2 energy is 2, but*

$$A(n) = 0, \quad D_{\text{bin}} = 0, \quad \mathbb{E}(f \mid \mathcal{G}) = 0.$$

Hence positive raw atomic energy does not survive determinant aggregation without a separate anti-cancellation or lower-modulus statement.

Example 4.7 (Unit-bin normalization can hide arbitrary mass). *Take two bins with aggregate values $A(1) = A(2) = 1$, but with fiber masses $M_1 = 1$ and $M_2 = M$. Then $D_{\text{bin}} = 2$, whereas*

$$\|\mathbb{E}(f \mid \mathcal{G})\|_2^2 = 1 + \frac{1}{M}.$$

Counting each bin with unit mass would return 2, but that is the norm in a different measure space.

5 The content projector returns the full shell

The small-content indicator in [\(14\)](#) admits an exact common-divisor Möbius expansion. At nonzero determinant frequency, this expansion can expose useful separated phases. At $r = 0$, its principal term is the original unrestricted amplitude.

5.1 Cumulative common-divisor inversion

For $C \geq 1$, define

$$\eta_{\leq C}(t) = \sum_{\substack{c|t \\ c \leq C}} \mu(t/c). \tag{42}$$

Lemma 5.1 (Exact cumulative projector). *For every positive integer G ,*

$$\boxed{\mathbf{1}_{G \leq C} = \sum_{t|G} \eta_{\leq C}(t).} \quad (43)$$

Moreover,

$$\eta_{\leq C}(1) = 1, \quad \boxed{\sum_{\substack{t|G \\ t>1}} \eta_{\leq C}(t) = -\mathbf{1}_{G>C}.} \quad (44)$$

Proof. Reversing the divisor sums gives

$$\sum_{t|G} \eta_{\leq C}(t) = \sum_{\substack{c|G \\ c \leq C}} \sum_{k|G/c} \mu(k).$$

The inner sum equals one precisely when $c = G$ and is zero otherwise. This proves (43). The value at $t = 1$ is immediate from (42). Subtracting it from (43) gives

$$\mathbf{1}_{G \leq C} - 1 = -\mathbf{1}_{G > C},$$

which is (44). $\square$

5.2 Physical reassembly at $r = 0$

Apply lemma 5.1 to

$$G = c(\omega) = \gcd(N_\alpha(j), N_\gamma(j))$$

for every literal atom $\omega = (\alpha, \gamma, j)$.

Theorem 5.2 (No content-projector centering). *At the physical cutoff $C = \lfloor J \rfloor$,*

$$\boxed{Z_X = \sum_{\omega \in \mathcal{I}_X} w_X(\omega) \sum_{t|c(\omega)} \eta_{\leq C}(t) = \mathcal{S}_{\text{full}}^{\text{sh}} - \mathcal{S}_{>C}^{\text{sh}}.} \quad (45)$$

More precisely:

$$t = 1 \text{ contribution} = \mathcal{S}_{\text{full}}^{\text{sh}}, \quad (46)$$

$$\text{sum of all } t > 1 \text{ contributions} = -\mathcal{S}_{>C}^{\text{sh}}. \quad (47)$$

Thus the content projector does not annihilate or center the distinguished zero mode.

Proof. Multiply (43) by $w_X(\omega)$ and sum over ω . The left side is the small-content sum, hence Z_X . Because $\eta_{\leq C}(1) = 1$, the $t = 1$ term sums every literal atom with its original signed coefficient, which is $\mathcal{S}_{\text{full}}^{\text{sh}}$. Equation (44) shows pointwise that the remaining divisor terms equal $-\mathbf{1}_{c(\omega) > C}$. Their sum is $-\mathcal{S}_{>C}^{\text{sh}}$. $\square$

Corollary 5.3 (The principal term contains the hard problem). *Using (21),*

$$Z_X = \mathcal{S}_{\text{full}}^{\text{sh}} + O(X^{o(1)} Q^2). \quad (48)$$

Therefore a bound $|Z_X| \ll X^{o(1)} Q^2$ is, at the same scale, a cancellation statement for the unrestricted full matched shell. It is not obtained by estimating only the nonprincipal projector terms.

Proof. Combine theorem 5.2 with (21). $\square$

Remark 5.4 (Exact reassembly versus termwise absolute bounds). The useful estimate for the $t > 1$ terms comes from their exact reassembly into the large-content shell. Replacing that signed sum by

$$\sum_{\omega} |w_X(\omega)| \sum_{\substack{t|c(\omega) \\ t>1}} |\eta_{\leq C}(t)|$$

can be much larger and does not create a bound for the $t = 1$ term. The fact that nontrivial divisors appear in the expansion is not, by itself, oscillation of the physical zero mode.

Remark 5.5 (Row calibration is a different operation). The calibrated row prefix subtracts a row-dependent term and produces two explicit drift legs; see [remark 2.1](#). Content inversion acts on the common target gcd. Neither identity says that $\sum_n A_C(n)$ vanishes, and the two operations must not be combined into an unrecorded “centered” coefficient.

6 The fixed-shift affine Möbius interface

The preceding sections show that neither Fourier centering, conditional expectation, nor content inversion removes Z_X . Opening an ultra divisor on one polarized leg instead exposes a two-sign affine Möbius correlation. We record the exact form and the costs that remain outside it.

6.1 Reflection of one physical column

Fix one opposite row and one of the two polarizations in [\(10\)](#). On the opened ultra leg, write a row as $m = \ell d$, and reflect a divisor u by

$$\ell dj + h_0 = us. \quad (49)$$

The actual row-gcd mask can be expanded after the opposite row has been fixed. On one divisor layer v , write

$$d = v\tilde{d}.$$

Primitive support gives

$$(s, \ell jv) = 1, \quad (s\ell jv, h_0) = 1. \quad (50)$$

Hence there is a unique residue $\tilde{d}_0 \pmod{s}$ satisfying

$$\ell jv\tilde{d}_0 + h_0 \equiv 0 \pmod{s}.$$

Put

$$\tilde{u}_0 = \frac{\ell jv\tilde{d}_0 + h_0}{s}, \quad a = \ell jv. \quad (51)$$

Every solution on this layer is

$$\tilde{d}_r = \tilde{d}_0 + sr, \quad u_r = \tilde{u}_0 + ar, \quad (52)$$

and the determinant is the original fixed shift:

$$\boxed{s\tilde{u}_0 - a\tilde{d}_0 = h_0 \neq 0.} \quad (53)$$

Proposition 6.1 (Literal affine-column interface). *For each fixed opposite row, polarization, and layer (ℓ, j, s, v) , the inner arithmetic sequence produced by the actual mask has the form*

$$\boxed{\mathcal{C}_{\ell,j,s,v}(I) = \sum_{r \in I} \mu(\tilde{d}_0 + sr) \mu(\tilde{u}_0 + ar) \mathbf{1}_{(v, (\tilde{d}_0 + sr)(\tilde{u}_0 + ar))=1} \rho(r) W(r),} \quad (54)$$

where:

- (a) $a = \ell jv$, $(a, s) = 1$, $(as, h_0) = 1$, and $s\tilde{u}_0 - a\tilde{d}_0 = h_0$;
- (b) I is the union of at most two intervals and

$$|I| \ll 1 + \frac{D}{sv}; \quad (55)$$

- (c) ρ is a bounded fixed-period factor and W retains the logarithmic and smooth physical weights;
- (d) the outside coefficient retains the opposite row, the row-gcd divisor coefficient, and the remaining physical factors;
- (e) summation of the divisor coefficients for a fixed opposite squarefree row costs $X^{o(1)}$, as does expansion of the displayed local coprimality condition.

The symmetric construction covers the second polarization.

Proof. This is the actual-mask column disintegration of TPC-33 [10], with the shift renamed h_0 . The congruence preceding (51) gives (52) and (53). Sampling a row interval of length $O(D/v)$ in a residue class modulo s gives (55). Exact sign fusion on squarefree support supplies the two Möbius factors. The row-gcd projector is expanded only after the opposite row is fixed, yielding the $X^{o(1)}$ divisor mass. Exchanging the row labels gives the other polarization. $\square$

Remark 6.2 (Column complexity is not matrix complexity). The $X^{o(1)}$ divisor cost in proposition 6.1 is a statement for one fixed opposite row. It is not an $X^{o(1)}$ projective decomposition of the complete joint row-pair mask. Returning the column estimates through the outer row sum remains part of the physical gate.

6.2 Integral normal form

The geometry in (53) has a useful coordinate-free description. Assume

$$a, s \in \mathbb{N}, \quad (a, s) = 1, \quad (as, h_0) = 1.$$

Choose $x, y \in \mathbb{Z}$ such that $sx - ay = 1$.

Proposition 6.3 (Fixed-determinant normal form). *Every integral solution of*

$$sU - aD = h_0 \tag{56}$$

has a unique representation

$$U_t = h_0x + at, \quad D_t = h_0y + st, \quad t \in \mathbb{Z}. \tag{57}$$

Furthermore,

$$\gcd(D_t, U_t) = \gcd(D_t, h_0) = \gcd(U_t, h_0) = \gcd(h_0, t). \tag{58}$$

Whenever $D_t, U_t > 0$, on the primitive layer $(t, h_0) = 1$,

$$\mu(D_t)\mu(U_t) = \mu(D_tU_t), \tag{59}$$

and the reducible quadratic D_tU_t has discriminant h_0^2 .

Proof. The matrix

$$\begin{pmatrix} x & a \\ y & s \end{pmatrix}$$

has determinant one. Applying it to $(h_0, t)^\top$ gives (57); applying its integral inverse proves uniqueness. A unimodular map preserves the ideal generated by the two coordinates, which gives $\gcd(D_t, U_t) = \gcd(h_0, t)$. The remaining gcd identities follow from $(as, h_0) = 1$. On the primitive layer the two forms are coprime, proving (59). Finally,

$$D_tU_t = ast^2 + h_0(sx + ay)t + h_0^2xy$$

has discriminant

$$h_0^2\{(sx + ay)^2 - 4asxy\} = h_0^2(sx - ay)^2 = h_0^2.$$

$\square$

Remark 6.4 (Structural identity, not cancellation). The broad normal-form class contains the classical adjacent correlation: with $a = s = h_0 = 1$, $x = 1$, and $y = 0$, one obtains $\mu(t)\mu(t+1)$. Therefore a theorem uniform over the entire normal-form class would contain an open Chowla case. This observation does not prove that the narrower physical subfamily in (54) is equivalent to that classical case.

6.3 The complete reassembly gate

Let θ collect the fixed opposite row, polarization, (ℓ, j, s, v) , interval component, residue factor, and smooth separation index. An exact use of [proposition 6.1](#) must have the form

$$Z_X = \sum_{\theta \in \Theta_X} W_{\theta, X} \mathcal{C}_{\theta, X} + E_{\text{content}}, \quad (60)$$

where $\mathcal{C}_{\theta, X}$ has [\(54\)](#) and

$$E_{\text{content}} = -\mathcal{S}_{>C}^{\text{sh}}. \quad (61)$$

This is the reassembly of the raw hard coefficient [\(14\)](#). The calibrated drift legs are not part of A_C , Z_X , or D_X . Equation [\(60\)](#) is an interface, not an estimate.

Proposition 6.5 (Sufficient uniform affine budget). *Suppose [\(60\)](#) has been verified with all native keys and both polarizations, and suppose that, for some $\eta_Z \geq 0$,*

$$\sum_{\theta \in \Theta_X} |W_{\theta, X} \mathcal{C}_{\theta, X}| \leq X^{-\eta_Z + o(1)} Q^2, \quad (62)$$

$$|E_{\text{content}}| \leq X^{-\eta_Z + o(1)} Q^2. \quad (63)$$

Then

$$|Z_X| \leq X^{-\eta_Z + o(1)} Q^2. \quad (64)$$

Proof. Apply the triangle inequality once, after the exact global reassembly [\(60\)](#), and use [equations \(62\) and \(63\)](#). $\square$

Remark 6.6 (Why the sufficient condition is demanding). The absolute sum in [\(62\)](#) is not asserted here. It may discard useful outer cancellation and can cost a fixed power. A stronger future argument may instead estimate the complete outer operator. Either route must still cover growing a, s , the $O(1)$ -length fibers allowed by [\(55\)](#), the actual mask, and the same fixed h_0 .

Remark 6.7 (The calibrated drift is downstream). TPC-32 writes the complete calibrated cutoff difference as the raw matched shell plus two explicitly bounded drift legs. Those legs must be restored when the raw estimate is returned to the calibrated physical packet, and their loss belongs in the complete downstream ledger. Inserting them into [\(60\)](#), however, would change its left side from the raw Z_X defined in [\(15\)](#) to a different calibrated zero mode. The two normalizations are therefore kept separate.

7 Quantifier audit of available correlation theorems

The affine sequence [\(54\)](#) resembles several established multiplicative-correlation problems. Resemblance of the summand is not enough: the averaging measure, coefficient range, selected shift, selected scale, interval length, and outer reassembly are theorem hypotheses.

7.1 Fixed-form logarithmic correlation

For fixed $a_1, a_2 \in \mathbb{N}$ and $b_1, b_2 \in \mathbb{Z}_{\geq 0}$ with $a_1 b_2 - a_2 b_1 \neq 0$, Tao proved the logarithmically averaged two-point Chowla estimate

$$\sum_{x/\omega(x) < n \leq x} \frac{\lambda(a_1 n + b_1) \lambda(a_2 n + b_2)}{n} = o(\log \omega(x)), \quad (65)$$

where $1 \leq \omega(x) \leq x$ and $\omega(x) \rightarrow \infty$ [\[5\]](#). The Elliott form of the theorem treats more general bounded multiplicative functions under its nonpretentiousness hypotheses and contains fixed-form Möbius analogues.

This is a genuine parity-breaking correlation theorem in its stated logarithmic setting. It does not, without additional arguments, provide the physical estimate because:

- (a) the slopes s and $a = \ell j v$, and the intercepts, vary polynomially with X ;

- (b) the required sum is dyadic or ordinary rather than weighted by $1/n$;
- (c) the packet contains many intervals, some of length $O(1)$;
- (d) fixed residue factors, growing auxiliary moduli, and the actual joint mask must be retained;
- (e) all columns and both polarizations must be reassembled with a quantitative fixed-power budget.

The entropy decrement argument is one mechanism in the proof of (65); it is not a black-box rule that deletes the logarithmic weight or supplies uniformity in growing affine coefficients.

7.2 Averaged shifts

Matomäki, Radziwiłł, and Tao proved, for fixed k and $1 \leq H = H(X) \leq X$ with $H(X) \rightarrow \infty$,

$$\sum_{h_1, \dots, h_k \leq H} \left| \sum_{n \leq X} \lambda(n + h_1) \cdots \lambda(n + h_k) \right| = o(H^k X). \quad (66)$$

See Matomäki et al. [4]. This ordinary, unweighted estimate is strong on average over shifts. It does not select one prescribed h_0 . An exceptional shift may be the same h_0 for every X while (66) remains true. Nor does the theorem directly carry the growing two-slope geometry and outer mask of (54).

7.3 Almost-all scales

Tao and Teräväinen proved unweighted cases of Chowla and Elliott at all scales outside a set of zero logarithmic density [6]. For fixed Liouville shifts, the stated Chowla cases include odd k and $k = 2$. In particular, their binary result is much closer to the ordinary fixed-shift correlation than (65). The remaining quantifier is nevertheless essential: a prescribed sparse family of physical scales may lie inside a zero-logarithmic-density exceptional set. A block theorem that must hold at every selected scale cannot discard that family without a separate scale-selection or interpolation argument.

7.4 Almost-all short intervals

The short-interval theory of Matomäki and Radziwiłł gives mean-value control for multiplicative functions on almost all short intervals [3]. Such results are a central input to later correlation theorems. A literal affine fiber, however, has a prescribed origin determined by $(\ell, j, s, v, \tilde{d}_0)$. An exceptional set of interval origins may contain that prescribed origin. In addition, the bound $|I| = O(1)$ permitted by (55) supplies no asymptotic short-interval parameter.

The same paper also proves a fixed-shift ordinary bound away from the trivial extremum for the mean of $\lambda(n)\lambda(n+1)$: its absolute value is at most $1 - \delta$ for some $\delta > 0$. This is an important nontrivial-correlation theorem, but it is not an $o(X)$ cancellation statement, a power-saving estimate, or a theorem uniform over the growing affine family in (54).

7.5 The quantifier table

Result	Native strength	Missing literal input
Tao 2016	Fixed nonproportional affine forms; logarithmic weight; general two-point Elliott form	Ordinary/dyadic weight, uniform growing coefficients, short-fiber and complete outer-reassembly bounds
Matomäki–Radziwiłł–Tao 2015	Ordinary correlations averaged over $h_1, \dots, h_k$	Selection of the prescribed h_0 , the physical two-slope family, and the literal mask
Tao–Teräväinen 2019	Unweighted correlations at scales outside a zero-log-density set	Every prescribed physical scale, growing affine coefficients, and a fixed-power reassembly rate
Matomäki–Radziwiłł 2016	Multiplicative mean values on almost all short intervals	The prescribed fiber origin, two-sign affine product, and $O(1)$ -length fibers

Table 1: The theorem may be strong in its native quantifiers while still failing to imply the literal fixed- h_0 packet estimate.

7.6 Elementary localization barriers

Proposition 7.1 (Averages do not select a prescribed member). *Let h_0 be fixed. A bound of the form*

$$\frac{1}{H} \sum_{h \leq H} |B_X(h)| = o(1)$$

does not logically imply $B_X(h_0) = o(1)$. Likewise, a statement holding outside a zero-logarithmic-density set of scales does not logically imply the statement along a prescribed sparse sequence of scales.

Proof. For the first assertion, set $B_X(h_0) = 1$ and $B_X(h) = 0$ for $h \neq h_0$. The average is $1/H = o(1)$, while the selected value is one. For the second, place every prescribed sparse scale in the exceptional set. For example, the set $\{2^k : k \geq 0\}$ has zero logarithmic density since

$$\frac{1}{\log X} \sum_{2^k \leq X} \frac{1}{2^k} \rightarrow 0.$$

The almost-all-scale assertion may therefore hold off that set while failing at every selected dyadic scale. $\square$

Remark 7.2 (Logical countermodel versus arithmetic counterexample). [Proposition 7.1](#) is a statement about quantifiers. It does not assert that the actual Möbius correlation is exceptional at h_0 or at dyadic scales. It shows only that an additional localization theorem is required before an averaged result can be invoked for the literal packet.

Corollary 7.3 (What a callable theorem must match). *To enter (60) without an unreturned average, an analytic theorem or a proved auxiliary transfer must explicitly match:*

- (i) *the same fixed h_0 ;*
- (ii) *the actual ranges of $a, s, \tilde{d}_0, \tilde{u}_0$;*
- (iii) *the ordinary physical interval and its weight;*
- (iv) *long and bounded-length fibers;*
- (v) *periodic, coprimality, and joint-mask factors;*
- (vi) *a uniform rate surviving the complete outer sum.*

Failure to match one item may still yield a valid frozen, averaged, or restricted-sector theorem, but not the literal Z_X bound.

8 A black-box log-to-unweighted counterexample

The mismatch between (65) and the ordinary physical sum is not a cosmetic choice of weight. We now show that no structure-free Tauberian principle can reverse it.

Definition 8.1 (Dyadic alternating sequence). For $n \geq 1$, define

$$c_n = (-1)^{\lfloor \log_2 n \rfloor}. \quad (67)$$

Thus $c_n = (-1)^k$ on $2^k \leq n < 2^{k+1}$.

Theorem 8.2 (Logarithmic cancellation without ordinary cancellation). *There is an absolute constant K_0 such that, for every $x \geq 2$ and $1 \leq \omega \leq x$,*

$$\left| \sum_{x/\omega < n \leq x} \frac{c_n}{n} \right| \leq K_0. \quad (68)$$

Consequently, for every function satisfying $1 \leq \omega(x) \leq x$ and $\omega(x) \rightarrow \infty$,

$$\sum_{x/\omega(x) < n \leq x} \frac{c_n}{n} = o(\log \omega(x)). \quad (69)$$

On the other hand, for $X_K = 2^{K+1} - 1$,

$$\sum_{n \leq X_K} c_n = \frac{1 - (-2)^{K+1}}{3}, \quad (70)$$

so

$$\left| \sum_{n \leq X_K} c_n \right| \asymp X_K. \quad (71)$$

Proof. Let

$$H_k = \sum_{2^k \leq n < 2^{k+1}} \frac{1}{n}.$$

Integral comparison gives

$$H_k = \log 2 + O(2^{-k}). \quad (72)$$

The interval $(x/\omega, x]$ meets at most two dyadic blocks partially. The harmonic mass of each partial block is $O(1)$. All remaining blocks form a consecutive range, say $k = m, \dots, M$, and their contribution is

$$\sum_{k=m}^M (-1)^k H_k = \log 2 \sum_{k=m}^M (-1)^k + O\left(\sum_{k=m}^M 2^{-k}\right) = O(1).$$

This proves the uniform bound (68). Dividing it by $\log \omega(x) \rightarrow \infty$ proves (69).

Every complete block $2^k \leq n < 2^{k+1}$ has 2^k terms. Therefore

$$\sum_{n \leq X_K} c_n = \sum_{k=0}^K (-1)^k 2^k = \frac{1 - (-2)^{K+1}}{3},$$

which proves equations (70) and (71). $\square$

Corollary 8.3 (No structure-free Tauberian black box). *For bounded sequences in general, an estimate of the form*

$$\sum_{x/\omega(x) < n \leq x} \frac{a_n}{n} = o(\log \omega(x)) \quad (73)$$

for every $1 \leq \omega(x) \leq x$ with $\omega(x) \rightarrow \infty$ does not imply $\sum_{n \leq X} a_n = o(X)$.

Proof. Take $a_n = c_n$ and apply [theorem 8.2](#). $\square$

Remark 8.4 (The precise scope of the counterexample). The sequence c_n is not multiplicative. Therefore [corollary 8.3](#) does not show that a conversion is impossible for Möbius or Liouville correlations. It shows that such a conversion must use additional structure not present in the logarithmic estimate alone. In particular, partial summation cannot be run backwards without quantitative information controlling the ordinary partial sums or their variation across scales.

Remark 8.5 (Direction of easy transfer). An ordinary estimate with sufficiently uniform control of partial sums can often be converted to a logarithmically weighted estimate by summation by parts. The counterexample concerns the reverse direction, which loses the dyadic block mass hidden by the $1/n$ weight.

9 Endpoint ledger and operational stop rules

The normalization barriers above matter because the physical route has a fixed endpoint budget. We record the two-exponent transfer without changing the hypotheses of TPC-81 and TPC-82 [\[11, 12\]](#). Throughout this section Z_X and D_X remain the raw hard quantities of [\(15\)](#). The calibrated drift is restored only after that raw gate and is charged separately in Λ_{phys} .

9.1 Zero mode and determinant energy

Assume $D_X > 0$ and define

$$\mathfrak{F}_{h_0}(X) = \frac{|Z_X|^2}{D_X}. \quad (74)$$

At the critical physical scales,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}. \quad (75)$$

Proposition 9.1 (Two-exponent transfer). *Suppose the literal coefficient at the same fixed h_0 and $C = \lfloor J \rfloor$ satisfies*

$$|Z_X| \leq X^{-\eta_Z+o(1)} Q^2, \quad (76)$$

$$D_X \geq X^{-\lambda_D+o(1)} Q^3 J^2. \quad (77)$$

Then

$$\boxed{\mathfrak{F}_{h_0}(X) \leq X^{1/400+\lambda_D-2\eta_Z+o(1)}}. \quad (78)$$

The recorded hypotheses imply the TPC-32 threshold

$$\mathfrak{F}_{h_0}(X) \leq X^{1/400+o(1)} \quad (79)$$

whenever

$$2\eta_Z \geq \lambda_D. \quad (80)$$

Proof. Divide the square of [\(76\)](#) by [\(77\)](#):

$$\mathfrak{F}_{h_0}(X) \leq X^{\lambda_D-2\eta_Z+o(1)} \frac{Q}{J^2}.$$

By [\(75\)](#),

$$\frac{Q}{J^2} = X^{(267-266)/400+o(1)} = X^{1/400+o(1)}.$$

This proves [\(78\)](#), and [\(80\)](#) gives [\(79\)](#). $\square$

Remark 9.2 (A sufficient ledger, not a converse). [Proposition 9.1](#) describes what the two recorded bounds imply. It does not claim that [\(80\)](#) is necessary for every possible proof of [\(79\)](#). Equality $2\eta_Z = \lambda_D$ reaches the conditional threshold with no fixed-power reserve.

9.2 The strict physical loss rule

Let Λ_{phys} denote the sum of all fixed-power losses actually incurred by the complete physical route, including any activation, localization, mask decomposition, dictionary, short-fiber, outer-reassembly, and remainder losses that are not already part of a baseline exponent.

Definition 9.3 (Strict endpoint rule). The route continues only under

$$\boxed{\Lambda_{\text{phys}} < \frac{1}{400}}. \quad (81)$$

If $\Lambda_{\text{phys}} = 1/400$ or $\Lambda_{\text{phys}} > 1/400$, the fixed-power route stops.

Remark 9.4 (The baseline is not a second allowance). The $1/400$ in (78) is the geometric baseline Q/J^2 . It is not another $1/400$ that may be spent after (81). Likewise, $2\eta_Z = \lambda_D$ supplies zero slack; it is not evidence of a positive reserve.

9.3 Route-II success and kill certificates

Definition 9.5 (Literal zero-mode success certificate). A proposed affine-correlation advance qualifies as a literal L2 input for (76) only if it verifies all of the following:

- (i) one exact identity returning the complete Z_X of (15), with the same fixed h_0 , $C = \lfloor J \rfloor$, and Fourier frequency $r = 0$;
- (ii) both polarizations, all matched channels, complete native keys, and the actual joint mask;
- (iii) the physical post-aggregation normalization, with no fiber-by-fiber unit renormalization;
- (iv) uniform estimates for growing affine coefficients, with long and $O(1)$ -length fibers separately charged;
- (v) no unreturned average over shifts, coefficients, interval origins, frequencies, or scales;
- (vi) if nonzero determinant frequencies are used in place of a direct $r = 0$ estimate, the complete coherent identity (32), including every exceptional nonzero mode, is returned;
- (vii) a complete fixed-power ledger satisfying (81).

Definition 9.6 (Kill or downgrade certificate). A calculation is downgraded to a frozen, averaged, restricted-sector, or negative-transfer result if any of the following occurs:

- (a) only one fixed column is controlled and the outer row/mask reassembly is not returned;
- (b) absolute outer summation costs an unbudgeted fixed power;
- (c) the invoked theorem fixes its slopes while the physical slopes grow with X ;
- (d) a positive proportion of the required mass lies in fibers with no asymptotic length parameter and no separate outer-reassembly or exact short-fiber estimate controls that sector;
- (e) the estimate is only logarithmic, averaged-shift, almost-all-origin, or almost-all-scale, with no proved localization returning the physical member;
- (f) the estimate is only generic- r , with no control of the exceptional nonzero modes and no coherent completion through (32);
- (g) conditional expectation replaces D_X by $\sum_n |A_C(n)|^2/M_n$;
- (h) the available $o(1)$ is not uniform enough to support a claimed fixed-power exponent;
- (i) the complete physical loss reaches or exceeds $1/400$.

Remark 9.7 (A downgrade can still be a theorem). [Definition 9.6](#) is a claim-label rule, not a reason to discard a correct result. A frozen logarithmic affine theorem, an averaged parameter theorem, a restricted-sector saving, or an exact no-go statement may be independently publishable. It must simply not be reported as the literal fixed- h_0 zero-mode estimate.

10 Claim boundary and conclusion

10.1 What has been proved

The paper proves the following unconditional statements.

- (1) For the actual TPC-32 determinant coefficient at $C = \lfloor J \rfloor$,

$$A_C(0) = 0 \quad \text{and} \quad \hat{A}_{C,q}(0) = Z_X$$

are different assertions: the first does not by itself make the second vanish.

- (2) Every no-wrap cyclic realization has the orthogonal decomposition

$$D_X = \|A_C^\circ\|_2^2 + \frac{|Z_X|^2}{q}.$$

Under the strong physical padding, it also has the coherent reassembly

$$Z_X = - \sum_{r \neq 0} \hat{A}_{C,q}(r).$$

- (3) Conditional expectation onto determinant fibers has norm $\sum_n |A_C(n)|^2/M_n$, preserves Z_X , and cannot be identified with the post-bin energy without fiber-mass control.
- (4) The cumulative content projector has full-shell $t = 1$ term and negative large-content $t > 1$ reassembly; it does not center the zero mode.
- (5) The opened physical column has a fixed-determinant affine Möbius interface, but its slopes grow, some fibers are bounded, and the complete joint-mask reassembly remains part of the gate.
- (6) Logarithmic cancellation does not imply ordinary cancellation for arbitrary bounded sequences; the conversion needs additional structure.

Items (1)–(4) are finite L0 identities, while item (6) is an explicit structure-free L0 countermodel. Their exact placement in the literal matched packet, together with the theorem-quantifier audit, is an L1 diagnosis. None is a new L2 Möbius-correlation estimate.

10.2 Explicit exclusions

This paper does not prove:

- (a) a bound of the form $|Z_X| \leq X^{-\eta_Z + o(1)} Q^2$ with a new $\eta_Z > 0$;
- (b) the physical determinant-energy lower bound for D_X ;
- (c) a uniform Chowla or Elliott theorem for the growing affine family in (54);
- (d) a full-block cover or the downstream uniform remainder bounds of the original route [7];
- (e) a fixed- h_0 Hardy–Littlewood asymptotic;
- (f) a breach of the general sieve parity barrier;
- (g) a prime-pair lower bound or the twin-prime conjecture.

No conclusion is specialized from a general fixed nonzero h_0 to $h_0 = 2$. The appearance of the two signs $\mu(D_r)\mu(U_r)$ identifies an arithmetic gate; it does not solve that gate. Sieve parity remains a genuine boundary [1, 2].

10.3 Conclusion

The distinguished zero mode is not hidden in the notation. It is the total mass of the literal determinant coefficient and, modulo the known large-content error, the complete matched shell. The three most immediate reorganizations have exact outcomes:

Fourier centering	→	subtracts $(Z_X/q)\mathbf{1}$,
coherent nonzero-frequency reassembly	→	returns $-\sum_{r \neq 0} \hat{A}_{C,q}(r) = Z_X$,
conditional expectation	→	preserves Z_X and inserts M_n^{-1} ,
content inversion	→	returns the full shell at $t = 1$.

None creates cancellation. The coherent Fourier identity does, however, identify a valid alternative route: control the full phase-sensitive sum of the nonzero modes, including every exceptional nonzero mode.

The surviving positive interfaces are the coherent Fourier reassembly (32) and the fixed-determinant affine correlation (54). The next arithmetic advance must either prove a uniform estimate that survives one of these literal reassemblies, isolate a rigorously controlled physical sector, or establish a sharper impossibility theorem for a proposed transfer. Whichever outcome occurs, it must retain the same h_0 , the actual $r = 0$, $C = \lfloor J \rfloor$, one global normalization, and the strict loss rule $\Lambda_{\text{phys}} < 1/400$.

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